1. Introduction and Project Objectives

SYNTHETIC TEST DATA. This report was generated to test ResearchPilot. Harbor Point Community College, its staff, and every figure below are fictional and must not be cited as real research.

The Harbor Point Community College Energy Storage Pilot added a battery energy storage system to the campus microgrid in January 2025. The microgrid already had an 800 kW rooftop solar array serving the Science Building and the Library. Without storage, most of the midday solar output was exported to the grid at a low price, while electricity was bought back at a high price during the evening peak.

The pilot had three objectives:

Objective 1: Store midday solar energy and use it during the evening peak, reducing grid purchases.

Objective 2: Keep the Library’s emergency lighting and network equipment running for at least four hours during a grid outage.

Objective 3: Measure the real efficiency, capacity, safety, and costs of the system, to decide whether to install storage at the college’s second campus.

The pilot ran for twelve months, from January to December 2025. Section 2 compares the battery options that were considered, Section 3 reports measured efficiency and capacity, Section 4 covers safety and cost, and Section 5 lists the limitations of this report.

2. Battery Options and Specifications

Three battery technologies were compared before purchase. Table 1 lists the example specifications supplied for each option. All three options were sized for 1,000 kWh of usable capacity.

Table 1. Example battery specifications (synthetic test data). Capacity is usable capacity, efficiency is round-trip efficiency, and price is in USD per kWh of capacity.

Option	Chemistry	Capacity	Efficiency	Cycle life	Price
A	Lithium iron phosphate	1,000 kWh	92%	6,000	USD 420
B	Nickel manganese cobalt	1,000 kWh	94%	3,500	USD 390
C	Vanadium redox flow	1,000 kWh	75%	15,000	USD 610

Option A, lithium iron phosphate (LFP), was selected. It was not the cheapest option, but it combined a cycle life of 6,000 cycles with a lower fire risk than Option B. Option B, nickel manganese cobalt (NMC), had the highest efficiency and the lowest price, but its cycle life of 3,500 cycles was much shorter and it was rejected mainly for safety reasons (see Section 4). Option C, a vanadium redox flow battery, had the longest cycle life, but its 75% round-trip efficiency would have wasted too much of the stored solar energy.

3. Efficiency and Capacity Results

Measured results for the selected LFP system (Option A) over the twelve-month pilot:
Round-trip efficiency: the average measured round-trip efficiency was 90.5%, slightly below the rated 92%. Most of the difference came from inverter losses and the energy used by the cooling system.
Energy throughput: the battery charged 361 MWh of midday solar energy and discharged 327 MWh during the evening peak (327 / 361 = 90.5%). This corresponds to 330 full equivalent cycles, an average of 0.9 cycles per day.
Capacity: the usable capacity was 1,000 kWh at the start of the pilot and 982 kWh after twelve months, a capacity fade of 1.8% in the first year.
Power: the maximum discharge power is 250 kW, so a full discharge takes about four hours.
Outage performance: during the only grid outage of the year, on 14 October 2025, lasting 2 hours and 30 minutes, the battery kept the Library’s emergency lighting and network equipment running without interruption, as required by Objective 2.

4. Safety and Cost Considerations

Safety: lithium iron phosphate cells have higher thermal stability than nickel manganese cobalt cells, which was the main reason for choosing Option A over Option B. The battery is housed in a separate container placed 3 metres from the Library, with gas detection and a clean-agent fire suppression system. Two minor alarms occurred during the pilot: a cooling-fan fault in March and a door-sensor fault in August. Neither involved overheating, and there were no safety incidents.
Cost: Table 2 shows the installed cost of the system.
Table 2. Installed cost (synthetic test data)

Cost item	USD
Battery modules (1,000 kWh x USD 420/kWh)	420,000
Installation, inverter, grid integration	100,000
Total installed cost	520,000

The measured savings in 2025 were USD 61,000, from lower evening electricity purchases and lower peak-demand charges. The simple payback period is therefore about 8.5 years (520,000 / 61,000). These figures do not include the annual maintenance contract (see Section 5).

5. Limitations and Next Steps

This report has several limitations that should be considered before using its results:
Short period: only twelve months of data were collected, so long-term capacity fade is uncertain.
Single site and mild year: the results come from one campus, and the 2025 winter was milder than average, so they may not transfer to the second campus.
Maintenance cost: the cost figures exclude the maintenance contract of USD 8,000 per year. Including it, the simple payback period would be about 9.8 years (520,000 / 53,000).
No environmental assessment: no life-cycle environmental assessment of the batteries was carried out.
Supplier figures: the specifications in Table 1 are example values from supplier quotes in this synthetic scenario, not independent laboratory tests.
Next steps: the college will collect a second year of data in 2026 before deciding whether to install the same system at its second campus.